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- **Symmetric Matrix**

  A symmetric matrix is a square matrix that is equal to its transpose. This means that for a matrix A, it holds that

  $$A = A^T.$$

  Symmetric matrices have several important properties:

  - Real Eigenvalues: All eigenvalues of a symmetric matrix are real numbers.
  - Orthogonal Eigenvectors: The eigenvectors corresponding to distinct eigenvalues are orthogonal to each other.

  Mathematically, if A is symmetric, then

  $$a_{ij} = a_{ji}$$

  for all elements of the matrix, where a_{ij} represents the element in the i-th row and j-th column.

- **Trace of a Matrix**

  The trace of a matrix, denoted as $\mathrm{tr}(A)$, is the sum of the diagonal elements of a square matrix. It has the following properties:

  - The trace is also equal to the sum of the eigenvalues of the matrix.

  For a matrix

  $$A = \begin{bmatrix} a_{11} & a_{12} & \cdots \\ a_{21} & a_{22} & \cdots \\ \vdots & \vdots & \ddots \end{bmatrix},$$

  the trace is given by:

  $$\mathrm{tr}(A) = a_{11} + a_{22} + \cdots + a_{nn}.$$

Problems

a) Given the matrix

$$A = \begin{bmatrix} 4 & 2 & 0 \\ 2 & 3 & 1 \\ 0 & 1 & 5 \end{bmatrix},$$

Verify that A is symmetric.

A matrix A is symmetric if $A = A^T$. The transpose of matrix A is:

$$A^T = \begin{bmatrix} 4 & 2 & 0 \\ 2 & 3 & 1 \\ 0 & 1 & 5 \end{bmatrix}.$$

Since $A = A^T$, the matrix A is indeed symmetric.

b) Given the matrix:

$$A = \begin{bmatrix} -3 & 1 \\ 1 & -3 \end{bmatrix}$$

To find the trace of the matrix A:

$$\text{tr}(A) = -3 + (-3) = -6.$$

So, the trace of the matrix A is -6.

Reference: Nicholson, W. K. (2006). Elementary Linear Algebra. ISBN 85-86804-92-4.